Backfilling And Double Writing 99
Separating Reads and Writes for Backfills 99
Specialized Tables For Backfills 100
Practicing Backfilling Techniques 101
Wrapping Up 102

6. **Optimizing Active Record** 105
Preferring Active Record Over SQL 105
Query Logs To Connect SQL to App Code 106
Common Active Record Problems 107
Use Eager Loading To Reduce Queries 110
Eager Loading With .includes() 110
Prefer Strict Loading Over Lazy Loading 111
Optimizing Individual Active Record Queries 112
Save A SELECT With A RETURNING 113
Bounding Query Results Using LIMIT 114
Advanced Query Support In Active Record 115
Using Common Table Expressions (CTE) 116
Introducing Database Views for Rideshare 118
Creating the Search Result Model With Scenic 120
Improving Performance With Materialized Views 121
Reducing Queries With Active Record Caches 122
Prepared Statements With Active Record 123
Eliminating Slow Count Queries With Counter Caches 124
Performing Aggregations In the Database 126
Reduced Object Allocations With SQL In Active Record 127
Wrapping Up 128

7. **Improving Query Performance** 129
Logging Slow Queries With Active Support Notifications 130
Capture Query Statistics In Your Database 131
Rideshare Query Statistics 132
Introducing PgHero As a Performance Dashboard 133
Analyzing Query Execution Plans 135
Finding Missing Indexes 138
Logging Slow Queries 139
Automatically Gathering Execution Plans 140
Performing Maintenance First 141
Interpreting Index Scan Execution Plan Info 142
EXPLAIN Scan Nodes and Bitmap Scans 143
Adding Query Boundaries With Filtering and LIMIT 144